

Extraction, Classification, Embeddings, And Validation

Empirical Methods — Week 13

Teaching deck draft

Week 13 question

- Once unstructured data become variables, how do we know what they measure?
- Validation is part of identification when the constructed variable enters the economics.
- The goal is not another extraction workflow.
- The goal is to defend benchmarks, diagnostics, robustness, transportability, and downstream interpretation.

Text, images, audio, video, and multimodal records all need the same discipline.

Validation as identification

$$\widehat{M}_i = g(U_i; \widehat{\theta})$$

Layer	Question
Construct	Does the measure match the economic object?
Benchmark	Do labels or ground truth represent that object?
Predictive	Does the model recover labels out of sample?
Subgroup	Does performance vary by group, place, time, or modality?
Downstream	Do remaining errors change descriptive or causal claims?

A strong downstream design cannot rescue a poorly defined measure.

Annotation and benchmark design

Choice	What to specify	Why it matters
Unit	document, image, clip, response	defines the estimand and error structure
Label	construct vs proxy	determines what the model learns
Annotators	experts, RAs, crowd, subjects	changes interpretation and quality
Rubric	rules, examples, edge cases	makes disagreement diagnosable
Adjudication	vote, expert review, tie-break	shapes benchmark stability
Holdout	untouched labeled sample	prevents tuning to the benchmark

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

Agreement is useful, but agreement on the wrong construct is still wrong.

Precision, recall, and F1

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Precision matters when false positives are costly.
- Recall matters when missed positives are costly.
- F1 balances the two, but it is not a policy loss function.
- The right threshold depends on the research use.

$$\mathbb{E}[Y \mid \hat{p} = p] = p$$

- A model can rank cases well and still give bad probabilities.
- Calibration matters for prevalence, targeting, thresholding, welfare calculations, and cost-benefit analysis.
- Check calibration overall and by subgroup.
- Recalibration may fix probabilities without fixing construct validity.

In applied economics, a score is often interpreted. Calibration tells us whether that interpretation is defensible.

Subgroup performance and fairness

$$\Pr(\hat{Y}_i \neq Y_i \mid G_i = g)$$

Setting	Failure mode	Economic consequence
Job postings	lower recall for service work	understated remote-work access
Satellite images	weak transfer to dense informal areas	distorted spatial inequality
Audio	device or identity shortcuts	tone measure confounds speaker traits
Video	high labeling burden and context loss	interaction score lacks construct validity

Fairness diagnostics are also identification diagnostics when group differences enter the claim.

Drift and transportability

- Vocabulary changes across firms, platforms, occupations, and time.
- Images change with sensors, seasons, resolution, and geography.
- Audio changes with microphones, rooms, speakers, and recording norms.
- Video changes with platform, consent, camera angle, and interaction setting.

Transport checks:

- hold out time, place, firm, occupation, annotator, or sensor;
- relabel in the target domain;
- report what breaks before using the measure downstream.

Embedding similarity and external validation

$$S_i(c) = \frac{e_i'c}{\|e_i\|\|c\|}$$

- Embeddings generalize beyond exact words or visual patterns.
- Similarity is not automatically exposure, feasibility, poverty, tone, or quality.
- Show nearest-neighbor audits, false positives, and false negatives.
- Compare to independent benchmarks not used in training.
- Test whether the same interpretation survives across groups and domains.

Measurement error in downstream economics

$$\tilde{X}_i = X_i + \nu_i$$

$$\text{plim } \hat{\beta}_{\tilde{X}} = \beta + \frac{\text{Cov}(\tilde{X}_i, \varepsilon_i - \beta \nu_i)}{\text{Var}(\tilde{X}_i)}$$

- Classical error can attenuate simple slopes.
- Differential error can change sign, subgroup gaps, and event-study dynamics.
- If measurement error is correlated with treatment, time, group, or outcomes, the economics changes.
- Robustness should include downstream estimates under alternative validated measures.

Validation examples from papers

Paper setting	Modality	Validation lesson
Hansen et al.	job-posting text	labels and model checks support remote-work comparisons
Haaland et al.	open-ended responses	coding, prompts, and human review define the object
Jean et al.	satellite imagery	image proxies need survey ground truth
Gorodnichenko et al.	audio	voice features need economic interpretation beyond acoustics
Blattman et al.	survey validation	qualitative evidence can reveal benchmark error

The shared standard: show what the measure captures and where it fails.

Reproduce

- labeled benchmark
- rater agreement
- dictionary baseline
- embedding similarity
- supervised score

Diagnose

- confusion matrix
- calibration bins
- subgroup metrics
- error audits
- downstream bias

Transfer

- new domain text
- vocabulary shift
- image proxy
- transport metrics
- redesign memo

The lab validates a constructed variable before interpreting it economically.

Takeaways

- Define the construct and benchmark before choosing the model.
- Report metrics that match the research use, not only headline accuracy.
- Audit subgroup performance, calibration, drift, and transfer.
- Treat human review and external validation as core evidence.
- Explain how residual measurement error affects downstream economics.